Kanghui Ning

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Kanghui Ning | kanghui-learning | Google Scholar | Kaggle

Storrs, CT - 06268, United States

Research Interests

My research focuses on building intelligent forecasting systems for real-world applications. I am currently working on pretraining **time-series foundation models**, with a focus on how data composition, synthetic data generation, and training strategies affect model generalization and forecasting performance. Beyond pretraining, I am particularly interested in **retrieval-augmented forecasting**, **test-time computation**, **LLM-empowered multi-modal time series analysis**, and **agentic approaches** that leverage external information and analytical tools to solve complex real-world forecasting tasks.

Education

- **University of Connecticut (UConn)** — *Aug 2024 – Present*
  Ph.D. in Computer Science and Engineering — Storrs, United States
  - GPA: 4.00/4.00
- **Huazhong University of Science and Technology (HUST)** — *Aug 2020 – Jun 2024*
  Master of Control Science and Engineering — Wuhan, China
  - GPA: 3.84/4.00
- **University of Electronic Science and Technology of China (UESTC)** — *Aug 2016 – Jun 2020*
  Bachelor of Automation (Honors Program) — Chengdu, China
  - GPA: 3.88/4.00

Publications

[1] K. Rasul, **K. Ning**, A. Schneider, and Y. Nevmyvaka, "The Arbitration Gap Is Mostly Noise: On Complementary Expertise in TSFMs," in *NeurIPS 2026 Workshop on Foundation Models for Temporal Systems (FMTS)*, 2026.

[2] K. Rasul, **K. Ning**, A. März, A. Mitra, A. Schneider, and Y. Nevmyvaka, "SWAFT: Sink-Augmented Sliding Windows for Time-Series Foundation Models," in *NeurIPS 2026 Workshop on Foundation Models for Temporal Systems (FMTS)*, 2026.

[3] **K. Ning**, M. Biloš, J. T. Wilson, Y. Zhang, K. Rasul, D. Song, A. Schneider, and Y. Nevmyvaka, "Test-Time Compute for Tabular Foundation Models: Mechanisms, Gains, and Limits," *arXiv preprint arXiv:2610.12005*, 2026. [arXiv] [GitHub]

[4] **K. Ning**, Y. Jiang, K. Rasul, A. Schneider, Y. Nevmyvaka, and D. Song, "TimeRouter: Efficient and Adaptive Routing of Time-Series Foundation Models," in *ICML 2026 Workshop on AI Forecasting*, 2026. [arXiv] [GitHub]

[5] Y. Jiang*, **K. Ning***, Z. Pan*, X. Shen, J. Ni, W. Yu, A. Schneider, H. Chen, Y. Nevmyvaka, and D. Song, "Multi-modal Time Series Analysis: A Tutorial and Survey," in *KDD 2025*, *Tutorial Track*, 2025. **Equal contribution* [arXiv] [GitHub]

[6] **K. Ning**, Z. Pan, Y. Liu, Y. Jiang, J. Y. Zhang, K. Rasul, A. Schneider, L. Ma, Y. Nevmyvaka, and D. Song, "TS-RAG: Retrieval-Augmented Generation based Time Series Foundation Models are Stronger Zero-Shot Forecaster," in *NeurIPS 2025*, 2025. [arXiv] [GitHub]

[7] **K. Ning**, Z. Pan, Y. Jiang, A. Schneider, Y. Nevmyvaka, and D. Song, "Towards Interpretable and Trustworthy Time Series Reasoning: A BlueSky Vision", in *ICDM 2025*, *BlueSky Track*, 2025. ***Best BlueSky Paper (3rd)*** [arXiv]

[8] Y. Jiang, **K. Ning**, W. Yu, H. Chen, and D. Song, "Learning Time-Frequency Graph Structures for Irregular Multivariate Time Series Forecasting", 2025 (Under review).

Research Experience

- **Morgan Stanley | Machine Learning Research** *Jun 2026 – Aug 2026*
  Research Intern, Mentor: Marin Biloš New York, United States
  - Conducted research on test-time compute for tabular foundation models, exploring inference-time strategies to improve predictive performance on tabular data.

- **University of Connecticut** *Aug 2024 – Present*
  Research Assistant, Advisor: Prof. Dongjin Song Storrs, United States
  - Conduct research on building reliable and interpretable AI systems for time series analysis, in collaboration with **Morgan Stanley**, focusing on **retrieval-augmented forecasting**, **multi-modal time series analysis**, and **time series reasoning systems**.

- **Ant Group | Intelligent Computing Division** *May 2024 – Aug 2024*
  Research Intern, Mentor: Yu Liu Hangzhou, China
  - Advanced LLMs for time series analysis, with a focus on retrieval-augmented mechanisms, and proposed a general framework for retrieval-augmented forecasting grounded in Time Series Foundation Models.

- **ByteDance | Corporate Strategy | Project Voyager** *Aug 2023 – Dec 2023*
  Research Intern, Mentor: Keats Shi, Yi He Beijing, China
  - Worked on RNA translation efficiency prediction, developing models and pipelines for accurate prediction and mRNA sequence design with high translation efficiency.

- **ByteDance | Applied Machine Learning | AI4Science** *Jun 2022 – Aug 2023*
  Research Intern, Mentor: Yuxuan Zhang Beijing, China
  - Conducted research on protein structure prediction and drug discovery, focusing on developing machine learning models for efficient and accurate prediction of protein side-chain packing and protein-ligand interactions.

Skills

- **Programming Languages:** Python, C/C++, Shell
- **ML Frameworks & Tools:** PyTorch, HuggingFace, DeepSpeed/FSDP, vLLM, Weights & Biases, Slurm, Docker, Git, Agentic coding tools (Claude Code, Codex)

Honors and Awards

- **Predoctoral Fellowship (Platinum)** *2026*
  School of Computing, University of Connecticut
- **Best BlueSky Paper (3rd)** *2025*
  ICDM 2025, BlueSky Track
- **Fellow, Entrepreneurship Fellowship Program** *2025*
  University of Connecticut
- **Scholarships (selected)** *2016 – 2025*
  *UConn CSE Graduate Fellowship; National Inspirational Scholarship; Outstanding Student Scholarship * 3; Academic Scholarship * 3; Bosch Scholarship*
- **Predoctoral Honorable Mention** *2025*
  School of Computing, University of Connecticut
- **Kaggle Expert** *2023*
  Top 1.4% among global competitors on Kaggle
- **Honorable Mention** *2019*
  Mathematical Contest in Modeling (MCM)
- **First Prize** *2018*
  Sichuan Province College Student Electronic Design Competition
- **First Prize** *2017*
  7th Ecovacs Intelligent Robot Competition, University of Electronic Science and Technology of China